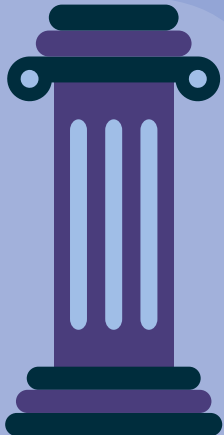


MA Final Project: Criminal Justice Dataset

Faith Sconing and Leela Addepalli



Description of Dataset

The National Crime Victimization Survey has collected annual data on personal and property crime since 1973.

This dataset is from 2023.



Description of Dataset

Address Record

Household address, year,
sample number, person ID
255,901 observations

Household Record

Same as address + land
ownership, usage, residents,
physical attributes etc.
254,071 observations

Person Record

Demographic information
275,343 observations

Incident Record

Pre, incident, post
information - perceived
cause, injuries, impact of
incident on work, health etc.
9324 observations



Research

Question

How does gender affect the likelihood of being a victim of violent personal crime?

Incidents Model

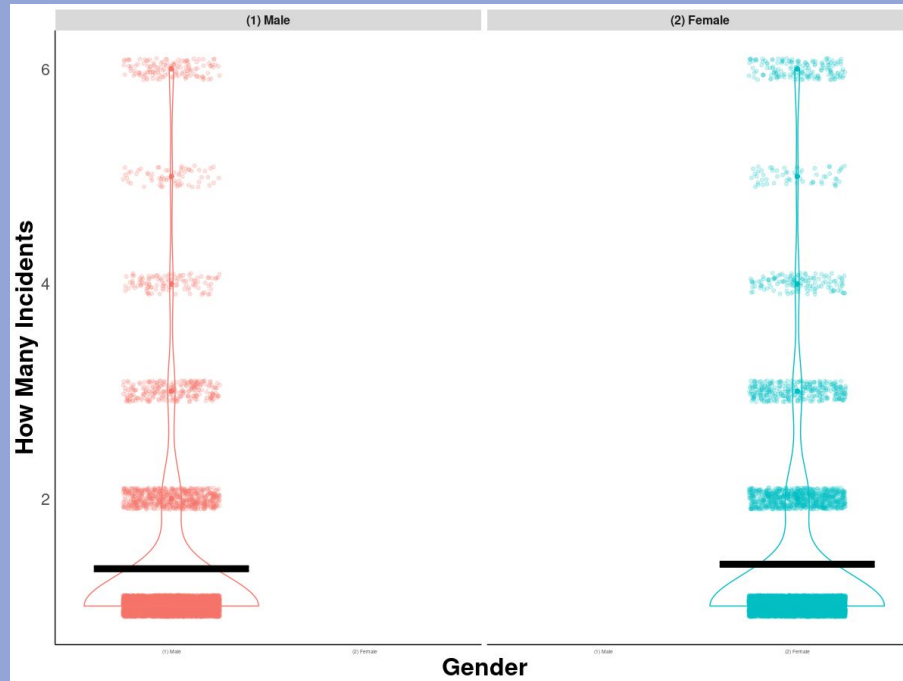
1. **Aggravated Battery** = Attempt with Weapon Present, Attempt with Knife, Attempt with Other Weapon, Attempt: Object Thrown, Attempt: Threat to Kill
2. **Battery/Other** = Attempt: Hit, Grab, Jump, Attempt: Other, Followed or Surrounded.
3. **Sexual Crime:** Threat of Rape, Threat of Sexual Assault, Sex Contact with Force

4. **Aggravated Battery:** Shot, Stabbed/Cut with Knife, Hit by Object held in hand, Hit by Thrown Object, Attack with Other Weapon, Threat and then Attack
5. **Battery:** Hit, slapped, knocked down, grabbed, held, tripped
6. **Sexual Crime:** Rape, Sexual Assault, Tried to Rape



Expert Graphs - Data comparison

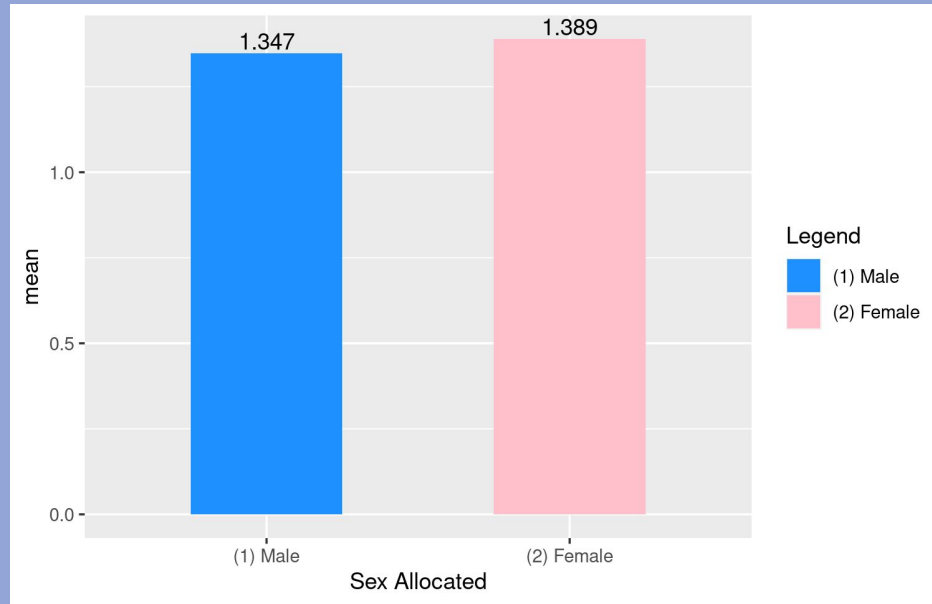
How many Incidents ~ Gender



Probability of
superiority = .49;
95% CI: [.48, .50]

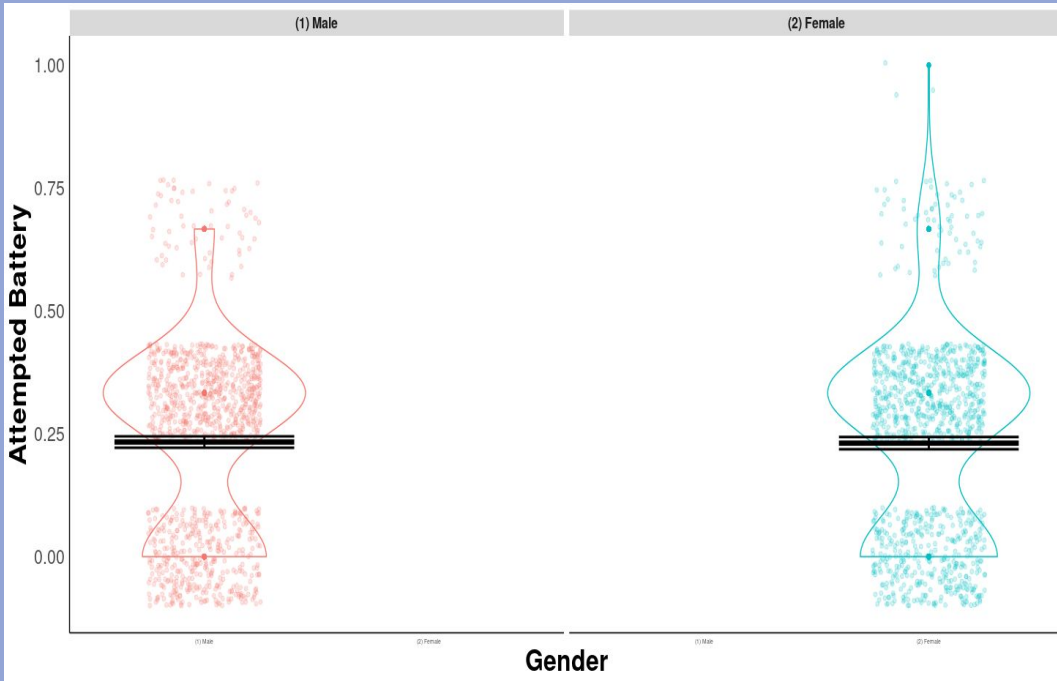
Laymen Graphs - Data comparison

How many Incidents ~ Gender



Expert Graphs - Data comparison

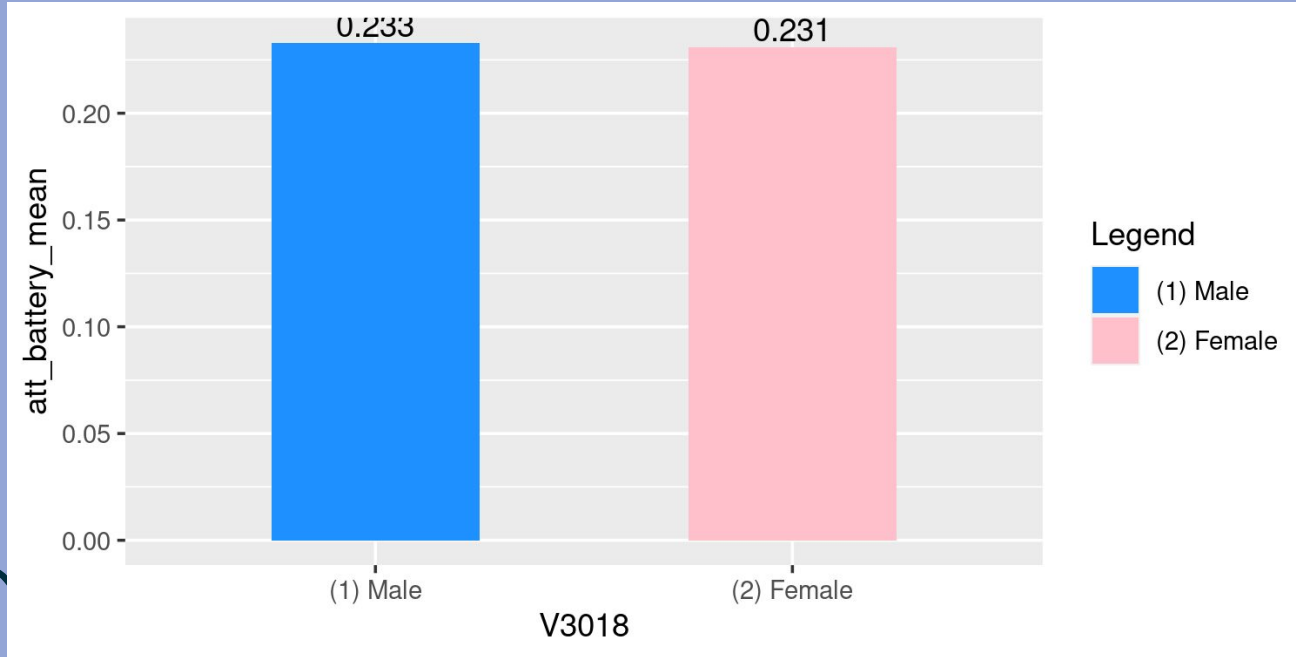
Attempted Battery ~ Gender



Probability of
superiority = .50;
95% CI: [.48, .53]

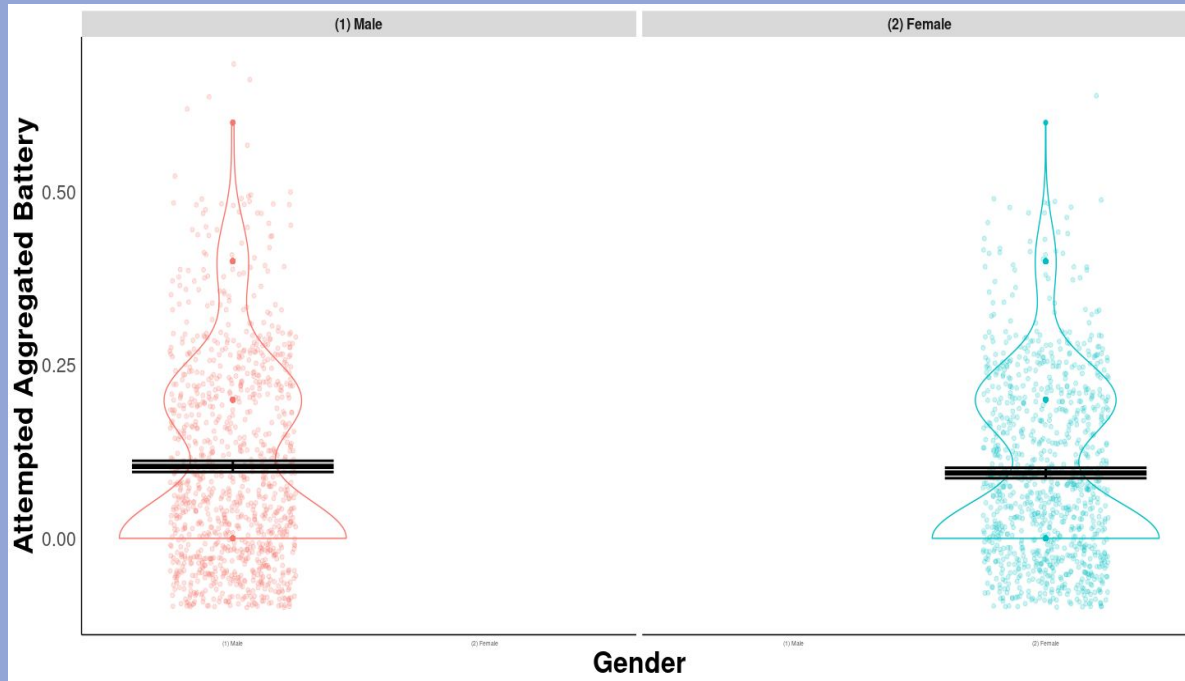
Laymen Graphs - Data comparison

Att. Battery ~ Gender



Expert Graphs - Data comparison

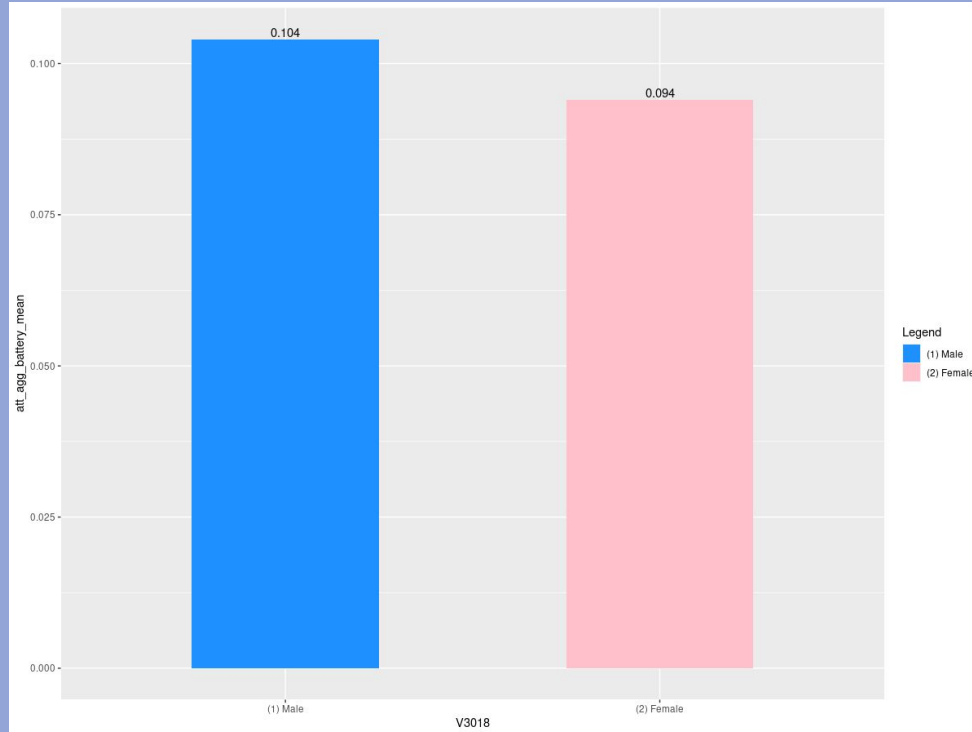
Attempted Agg. Battery ~ Gender



Probability of
superiority = .52;
95% CI: [.50, .55]

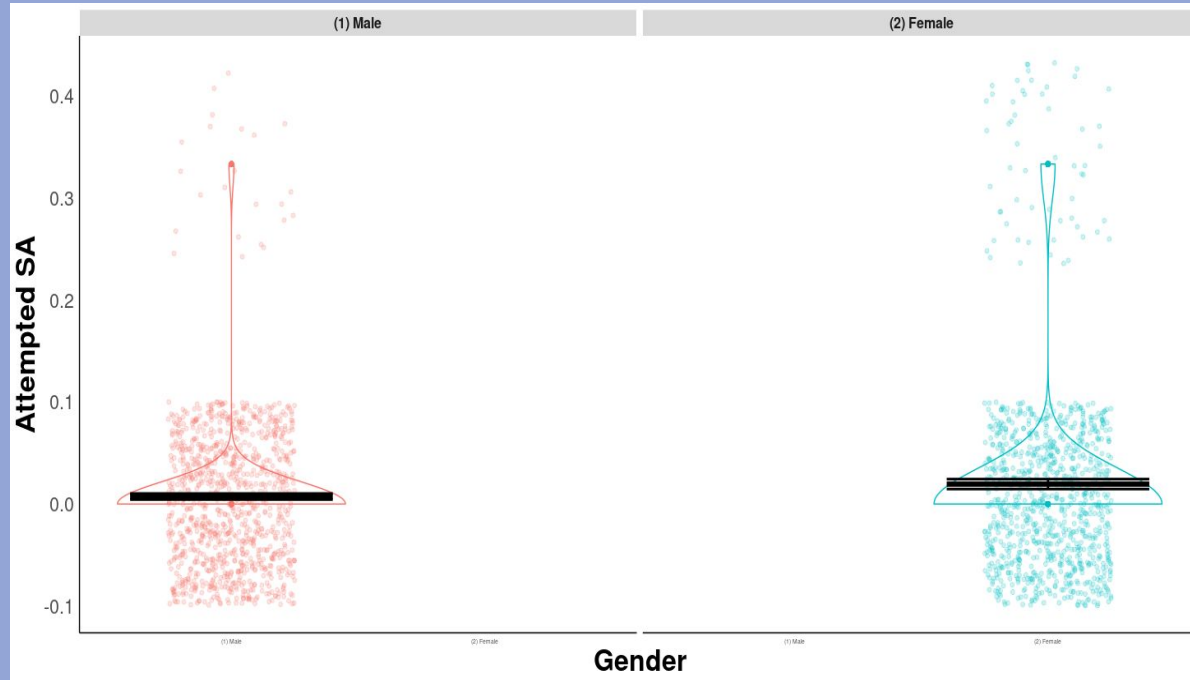
Laymen Graphs - Data comparison

Att. Agg. Battery ~ Gender



Expert Graphs - Data comparison

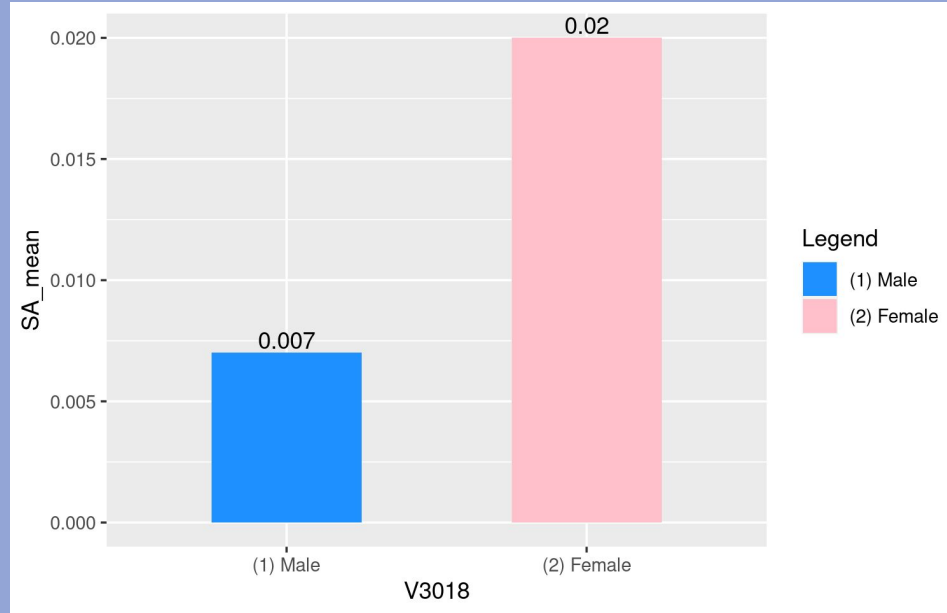
Attempted SA ~ Gender



Probability of
superiority = .45;
95% CI: [.42, .47]

Laymen Graphs - Data comparison

Att. SA ~ Gender



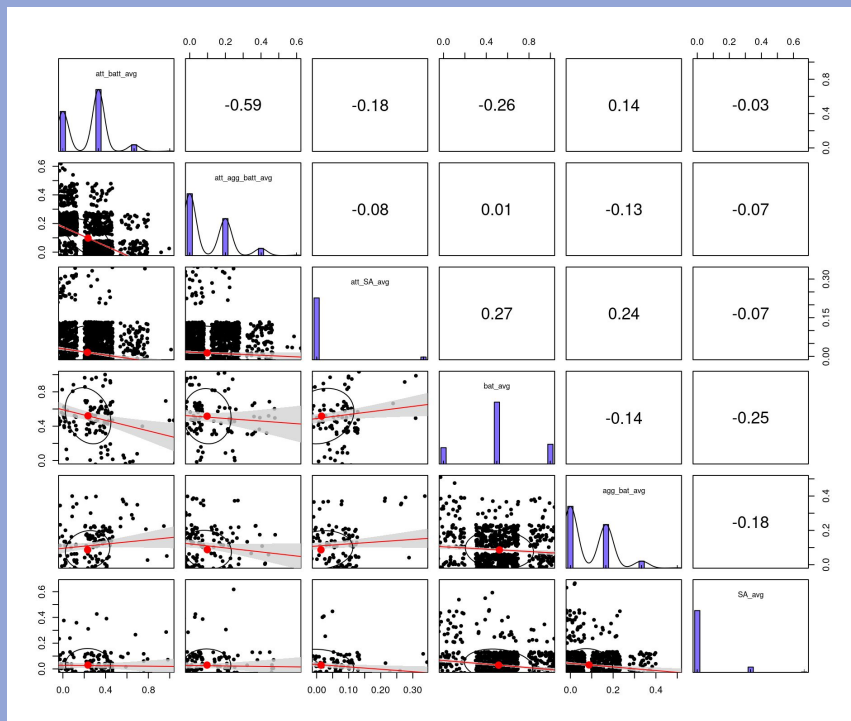


Guiding Question

What is the correlation
between attempts of violent
crime and actual occurrence
of violent crime?



Multiple Regression Results



Battery ~
Attempts

$R^2 = .111$, att_SA

Agg. Bat
~

Attempts

$R = .113$, att_SA

SA ~
Attempts

$R^2 = 0.036$, att_bat
(neg)

EFA Proposed Model

Attempted Violent Crime

1. **Aggravated Battery** = Attempt with Weapon Present, Attempt with Knife, Attempt with Other Weapon, Attempt: Object Thrown, Attempt: Threat to Kill
2. **Battery/Other** = Attempt: Hit, Grab, Jump, Attempt: Other, Followed or Surrounded.
3. **Sexual Crime**: Threat of Rape, Threat of Sexual Assault, Sex Contact with Force

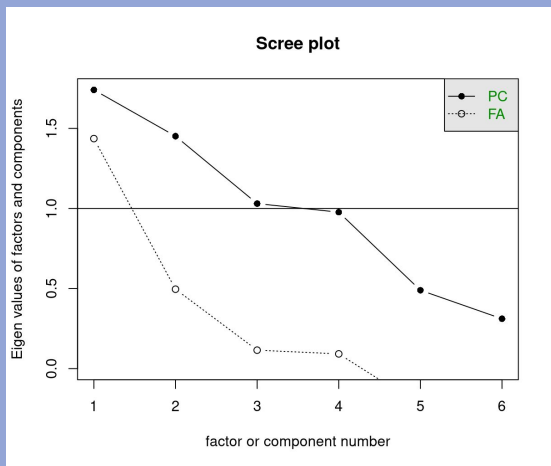


Actual Violent Crime

1. **Aggravated Battery**: Shot, Stabbed/Cut with Knife, Hit by Object held in hand, Hit by Thrown Object, Attack with Other Weapon, Threat and then Attack
2. **Battery**: Hit, slapped, knocked down, grabbed, held, tripped
3. **Sexual Crime**: Rape, Sexual Assault, Tried to Rape



EFA comparison



Statistics by number of factors

vss1	vss2	map	dof	chisq	prob	sqresid	fit	RMSEA	BIC	SABIC	complex
0.35	0.00	0.084	9	1.2e+03	2.5e-246	4.9	0.35	0.21	1099	1128	1.0
0.51	0.57	0.140	4	6.1e+02	1.4e-131	3.2	0.57	0.23	582	595	1.3
0.65	0.73	0.228	0	2.3e+02	NA	2.0	0.73	NA	NA	NA	1.3
0.69	0.81	0.434	-3	1.9e-10	NA	1.1	0.85	NA	NA	NA	1.4
0.69	0.79	1.000	-5	0.0e+00	NA	1.1	0.86	NA	NA	NA	1.5
0.69	0.79	NA	-6	0.0e+00	NA	1.1	0.86	NA	NA	NA	1.5

EFA Comparison

1 Factor Model

- RMSEA = .212, [.202, .233]

2 Factor Model

- RMSEA = .231 [.216, .246]

On comparing 90% CIs to measure goodness of fit:

$$.233 - .202 = .031 \times 2 = .062$$

.246 - .216 = .040, which is not $< .062$, therefore 1 factor model is not a reliably better fit than 2 factor.

Note - nfactors > 2 did not yield any result at all (no RMSEA)



Conclusions



Gender and Crime

Total number of incidents almost same for both Male and Female

Women more likely to be sexually assaulted but this difference is not reliable.

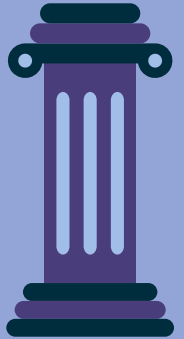


Attempt vs. Actual

Attempts of violent crime have a small correlation with actual battery and aggravated battery, but negatively correlated with sexual assault

Dimensions in Study

2 factor model chosen, EFA inconclusive due to data selection and cleaning procedure





Implications



- 
- 
- The way variables have been coded imply a difference in how different crimes are coded/charged in practice
 - Data highlights a well-known problem of the stigma and limited support offered to victims of sexual assault of either gender.

Ethical Reflection



1. Data Extraction

- Snapshot of data - 2023
- Visual analysis - limited time
- Other demographic variables not considered in study

2. Data Cleaning/Analysis

- NA data - lost almost 200,000 observations
 - Demographic analysis of missing data pending.
- Race - “mixed”
- EFA results were inconclusive
 - Potential reason: communalities





Thank You

Any questions, comments or feedback?

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